

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I

WEEK 12: LECTURE NOTE

GRAPH OF THE TANGENT FUNCTION: If $f(x) = \tan x$, recall that

1. The function f is odd. That is, $f(-x) = \tan(-x) = -\tan x = -f(x)$ for any $x \neq n\pi + \frac{\pi}{2}$. Consequently, the graph of $y = \tan x$ is symmetric with respect to the origin.
2. Domain of $f = \mathbb{R} \setminus \{n\pi + \frac{\pi}{2} : n \text{ is an integer}\}$.
3. Range of $f = \mathbb{R}$.
4. x-intercepts: $(n\pi, 0)$ for each integer n
5. y-intercept: $(0, 0)$
6. Vertical asymptotes: For each integer n the line $x = n\pi + \frac{\pi}{2}$ is a vertical asymptote.

Let us study the nature of the tangent function on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$. Use a calculator in radian mode to justify the values.

The following table gives you hint about the nature of $f(x) = \tan x$ as you are getting very close and close to $-\frac{\pi}{2}$ from the right side.

x	$-\frac{\pi}{2}$	-1.570796	-1.57079	-1.5707	-1.57	-1.5	0
$\tan x$	Undefined	-37,320,539	-158,058	-10,381	-1,256	-14	0

The following table gives you hint about the nature of $f(x) = \tan x$ as you are getting very close and close to $-\frac{\pi}{2}$ from the left side.

x	0	1.5	1.57	1.5707	1.57079	1.570796	$\frac{\pi}{2}$
$\tan x$	0	14	1,256	10,381	158,058	37,320,539	Undefined

As indicated in the tables,

1. $f(x) = \tan x$ decreases without bound as x approaches $-\frac{\pi}{2}$ from the right.
2. $f(x) = \tan x$ increases without bound as x approaches $\frac{\pi}{2}$ from the left.

So, the graph of $f(x) = \tan x$ has vertical asymptotes at

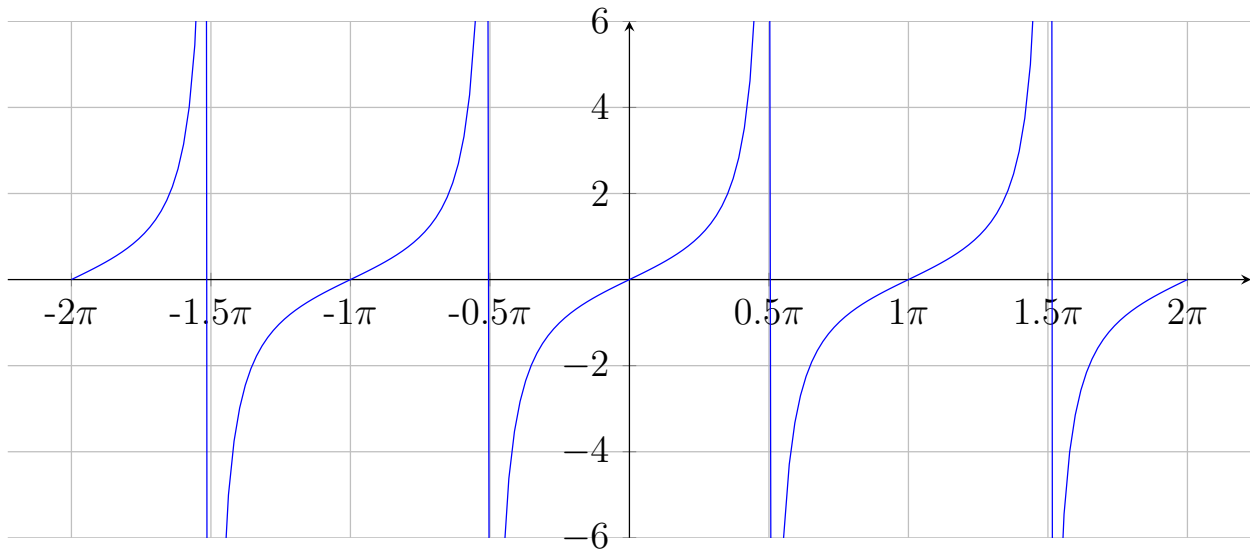
$$x = -\frac{\pi}{2} \text{ and } x = \frac{\pi}{2}.$$

Moreover, because the period of the tangent function is π , vertical asymptotes also occur at

$$x = n\pi + \frac{\pi}{2}, \quad n \text{ is an integer.}$$

With all the information from the last two slides the graph of the tangent function is given below.

GRAPH OF $f(x) = \tan x$

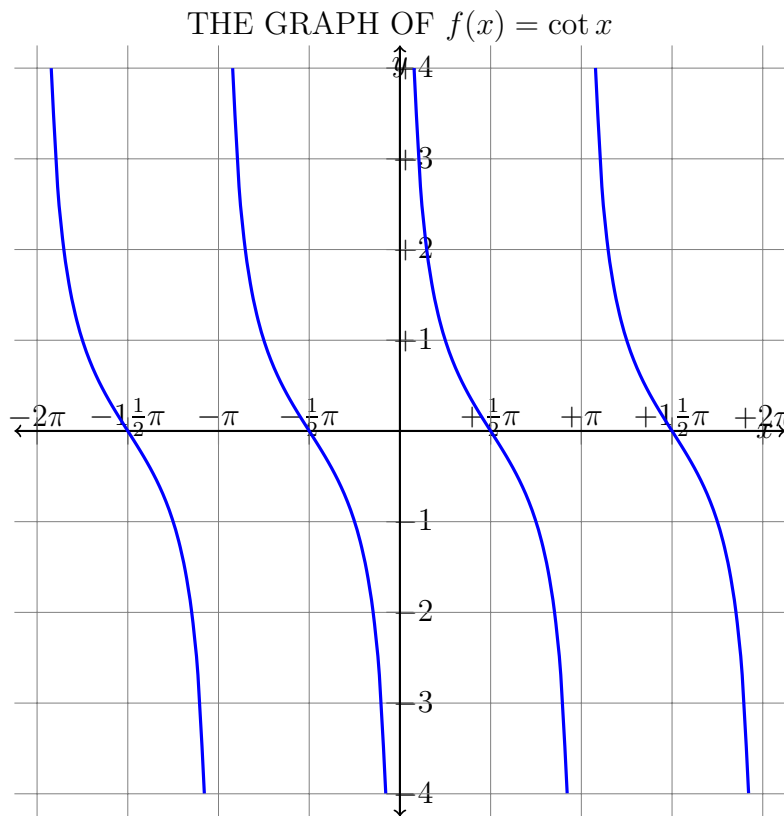


GRAPH OF THE COTANGENT FUNCTION

Verify the following properties of the cotangent function $f(x) = \cot x$.

1. Domain of $f = \mathbb{R} \setminus \{n\pi : n \text{ is an integer}\}$.
2. One cycle occurs between 0 and π .
3. Range of $f = (-\infty, \infty)$.
4. Vertical asymptotes: $x = n\pi$ for each integer n . The asymptote that occurs at π repeats every π units.
5. f is periodic with period: π .
6. The x-intercepts of the graph of $y = \tan x$ are the asymptotes of the graph of $y = \cot x$.
7. The asymptotes of the graph of $y = \tan x$ are the x-intercepts of the graph of $y = \cot x$.
8. The graphs of $y = \tan x$ and $y = \cot x$ have the same x-values for y-values of 1 and -1.
9. The graphs of $y = \tan x$ and $y = \cot x$ "face" in opposite directions.

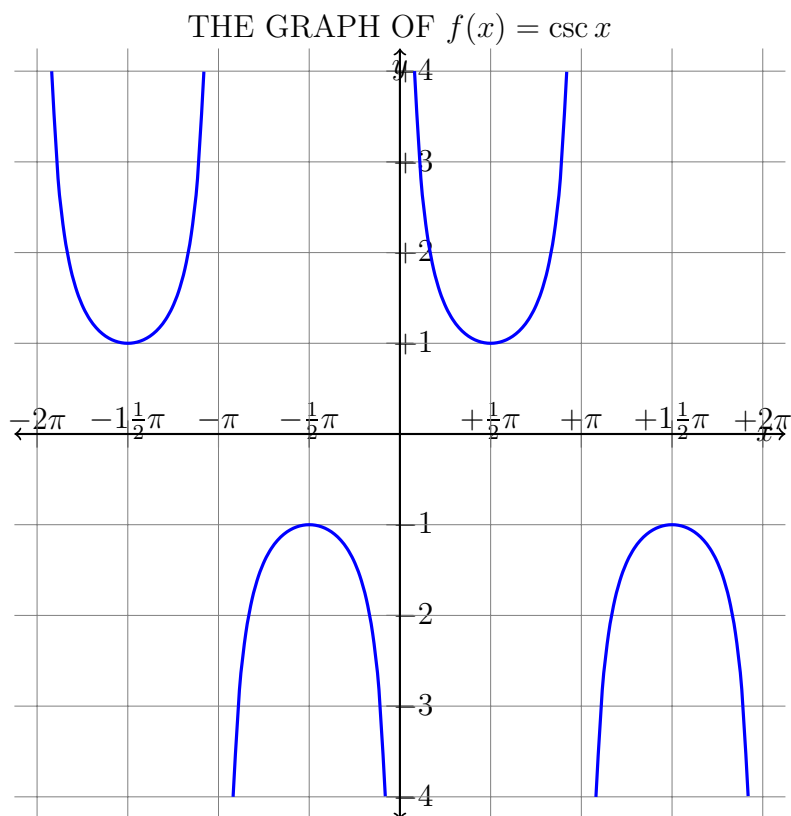
See the graph given below and compare it with the graph of the tangent function.



GRAPH OF THE COSECANT FUNCTION

Verify the following properties of the cotangent function $f(x) = \csc x$.

1. Domain of $f = \mathbb{R} \setminus \{n\pi : n \text{ is an integer}\}$.
2. Range of $f = (-\infty, -1] \cup [1, \infty)$.
3. Vertical asymptotes: $x = n\pi$ for each integer n . The asymptote that occurs at 0 repeats every π units.
4. f is 2π -periodic.
5. No x-intercepts
6. The maximum values of $y = \sin x$ are minimum values of the positive sections of $y = \csc x$.
7. The minimum values of $y = \sin x$ are the maximum values of the negative sections of $y = \csc x$.
8. The "U" shapes of the cosecant graph are tangent to its reciprocal function, sine, at sine's max and min locations.
9. Odd function. Symmetric with respect to the origin.



GRAPH OF THE SECANT FUNCTION

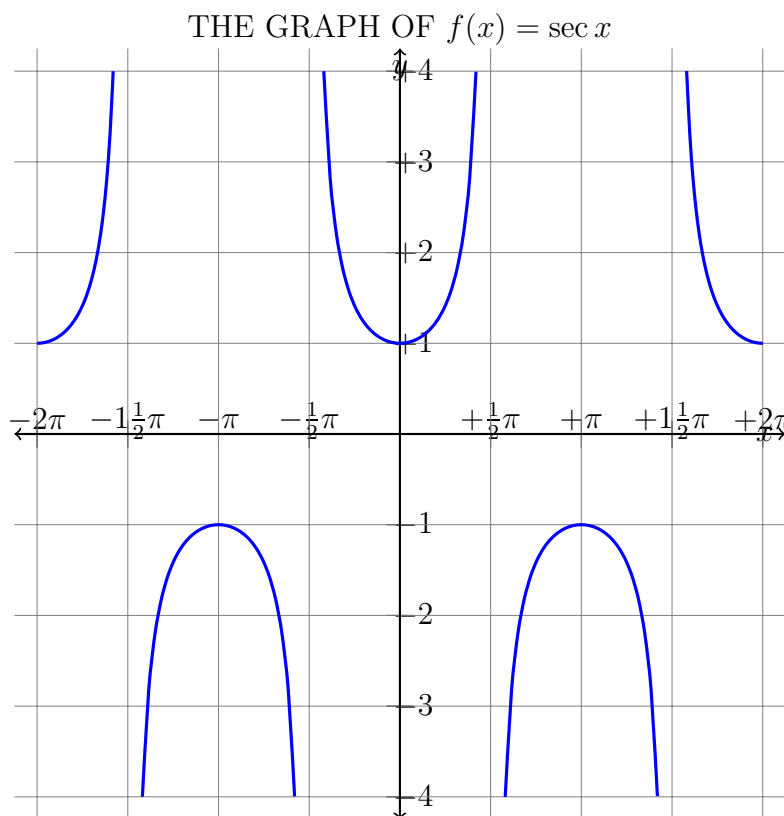
Check the following details for the cotangent function $f(x) = \sec x$.

1. Domain of $f = \mathbb{R} \setminus \{n\pi + \frac{\pi}{2} : n \text{ is an integer}\}$.
2. Range of $f = (-\infty, -1] \cup [1, \infty)$.
3. There are vertical asymptotes. The asymptote that occurs at $\frac{\pi}{2}$ repeats every π units. The vertical lines

$$x = n\pi + \frac{\pi}{2}$$

are vertical asymptotes of the graph of f .

4. f is 2π -periodic.
5. The maximum values of $y = \cos x$ are minimum values of the positive sections of $y = \sec x$. The minimum values of $y = \cos x$ are the maximum values of the negative sections of $y = \sec x$.
6. The x-intercepts of $y = \cos x$ are the asymptotes for $y = \sec x$.
7. The "U" shapes of the secant graph are tangent to its reciprocal function, cosine, at cosine's maximum and minimum locations.
8. Even function. Symmetric with respect to y-axis.



READING ASSIGNMENT: Recall the following rigid and non-rigid transformations discussed earlier. Try to analyze these transformations for trigonometric functions. (a , b , and c are positive numbers)

1. Horizontal shift: $h(x) = \sin(x - c)$ and $h(x) = \sin(x + c)$
2. Vertical shift: $h(x) = \sin x + c$ and $h(x) = \sin x - c$
3. Reflection about the x-axis: $h(x) = -\sin x$
4. Reflection about the y-axis: $h(x) = \sin(-x)$
5. Horizontal shrink (or Horizontal expand): $h(x) = \sin(bx)$
6. Vertical shrink (or Vertical expand): $h(x) = a \sin x$

In general analyze the properties of graphs of functions

$$h(x) = a \sin(bx + c) \text{ and } k(x) = a \cos(bx + c)$$

by taking the parent functions $f(x) = \sin x$ and $g(x) = \cos x$.

INVERSE TRIGONOMETRIC FUNCTIONS

Recall that for a function to have an inverse function, it must be one-to-one. That is, it must pass the **Horizontal Line Test**. By restricting the domain of a trigonometric function it is possible to obtain a one-to-one function; so that it is invertible in that restricted

domain. We suffice to the functions the phrase "arc" to mean inverse of that trigonometric function. The following table gives you domain and range of inverse trigonometric functions.

Function	Domain	Range
$y = \arcsin x \Leftrightarrow x = \sin y$	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
$y = \arccos x \Leftrightarrow x = \cos y$	$-1 \leq x \leq 1$	$0 \leq y \leq \pi$
$y = \arctan x \Leftrightarrow x = \tan y$	$-\infty < x < \infty$	$-\frac{\pi}{2} < y < \frac{\pi}{2}$
$y = \operatorname{arccot} x \Leftrightarrow x = \cot y$	$-\infty < x < \infty$	$0 < y < \pi$
$y = \operatorname{arcsec} x \Leftrightarrow x = \sec y$	$x \in (-\infty, -1] \cup [1, \infty)$	$y \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
$y = \operatorname{arccsc} x \Leftrightarrow x = \csc y$	$x \in (-\infty, -1] \cup [1, \infty)$	$y \in [-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

Sketch graphs of these functions as reflection along the line $y = x$ of the graphs of six trigonometric functions

